

3.4 Unit 4 INFO4 Coursework: Practical Issues Involved in the Use of ICT in the Digital World

This unit provides students with the opportunity to complete a substantial project involving the production of an ICT-related system over an extended period of time. In so doing, students will enhance their transferable practical skills.

The unit is designed to be taught alongside or after Unit 3 and topics covered in Unit 3 may provide the stimulus for work for this Unit 4, but the centre or student can explore new areas of ICT if they wish.

Possible projects

In addition to those given below, AQA will provide guidance on task setting in the Teacher Resource Bank. Annual Standardisation Meetings will also be provided for the purpose of giving guidance to teachers in the conduct of coursework and its marking.

There is the scope for work to be tailored to the individual needs of learners and for a wide range of ICT-related systems to be created. It is also important for each candidate's project to have a level of complexity and demand appropriate to the second year of an A Level.

The following may help to illustrate the range of possible projects that could be undertaken, and how groups of students could work together (see below), whilst still producing an individual project report that has their own work clearly identified.

Students may, for example, produce:

- a software solution such as an e-commerce or multimedia system

- a training system, including training materials for a client. This could, for example, be for a course for someone working from home
- a user support system, such as for a user help desk in a company or school/college, or a fault logging system
- a system for ensuring the security of an organisation's ICT systems. An idea here would be to formulate a policy to specify appropriate use of a company or a school's/college's laptop computers and other mobile devices, or a database to record usage
- a system for communication within an organisation: for example, for how schools could use technology to communicate with students within the school or a communal diary system
- a system for evaluating new software to be purchased or for a new system to be installed, including hardware, software, communications, consumables and services
- a backup and recovery system and a disaster recovery plan for an organisation
- a system for managing relationships with customers.

More suggestions are given later in this section along with ideas on how to enable groups of students to work together.

Working together

It is intended that this unit will provide opportunities for students to become involved in *real or realistic* situations in organisations where they can apply their skills, knowledge and understanding of ICT to solve problems for a client. An organisation is defined as any group of people who use ICT to achieve their aims and objectives. Examples of an organisation include a local company, a retailer, a school or college, a charity, a club or society, or just a group of people coming together to organise an event.

It is quite acceptable for students to work in teams for the same client, if that is what is required for the work being undertaken. Students at this level are expected to work, in some part, collaboratively with others. The team could consist of two or more students working with one or more members from the client organisation. This approach is encouraged if it allows students to become involved in larger more realistic projects where each student can work on part of a system.

Using software

The project should involve the students in the use of applications software for a variety of purposes and should also introduce students to the fundamentals of project management and, although there is no requirement to use dedicated project management software, students may wish to so that they can better manage their own project or their part in a larger project.

Students will have used a variety of applications software during their study of Unit 1; Unit 4 provides students with the opportunity to enhance their skills or develop new ones.

Students will be expected to make full use of the functionality available in word processing software for the production of the project report and, if required, to enable collaborative working.

Skills

The skills set down in the QCA Subject Criteria for GCE ICT are those necessary to undertake an ICT-related project: thus, as part of this unit, students will have the opportunity to enhance their transferable skills in:

- investigation
- analysis
- definition of requirements
- design of effective solutions to meet a requirements specification, including the methods to be used for testing and installation
- selection and use of appropriate technologies
- implementation

- testing
- documentation
- evaluation of solutions and their own performance.

The project should encourage students to think creatively, innovatively, analytically, logically and critically as they consider and suggest alternative approaches, investigate and discuss possibilities, and recommend actions to be taken.

Students will produce a project report which will be assessed internally by the centre, to the criteria provided in this specification. The centre's assessment will then be moderated by AQA. For this reason, although collaborative working is encouraged, it is *imperative* that assessors are able to identify each student's own contribution, and each student must produce their own project report.

Further project ideas

The following may help to illustrate the range of possible projects that could be undertaken, and how groups of students could work together, whilst still producing an individual project report that has their own work clearly identified.

- **Website, intranet site or extranet site**

For any organisation (or event), students could produce parts of an intranet for different departments within an organisation, working collaboratively to establish house styles and consistency of end-product, and in sharing their skills in the use of the variety of applications software needed to create the end-product.

- **Training materials**

Training materials could be developed for a course within the centre or for an external organisation. Materials could take a variety of formats and be created using a range of software. Individual students could work on one aspect or one format, whilst the whole group has to adopt consistent styles and approaches to development. For example, training could be needed in:

- a new piece of software
- using new laptops, PDAs or other portable devices
- using loan items such as digital cameras or recording equipment
- a new security policy
- a backup and recovery strategy
- health and safety legislation or guidelines
- environmental policies, such as disposal of computer equipment.

Other students could perhaps develop a system for storing, cataloguing or loaning out training materials that have been created.

- **User support and fault logging**

A group of students could each work with a different type of user, or a different department of a school or other organisation. Materials required might be similar in purpose, but need to be produced so as to best suit the individual types of user, or some may have specialist needs which need supporting. Recording and storage of requests for maintenance, error reporting or fault logging could be built in.

- **Managing relationships with customers**

This could involve a system for the collection of customer feedback via a website and a back office system that can collect, store and classify responses, plus a system that results can be fed into for statistical analysis. Individual students could take on one of the elements, where collaboration would be essential to ensure effective data transfer, correctness and accuracy.

- **Multimedia solutions**

These can be used in a variety of situations. They can be extremely large systems and frequently need to be broken down into manageable chunks. Separate students could work on parts of an overall solution. This could include producing an interactive map, where subject specialisms or special interests could determine with what areas students became involved. For example, one student could prepare and incorporate information on areas of interest for young adults, another on sporting venues, a third on flora and fauna. This could be produced for the local tourist board or town council.

Additional information and examples are available in the Teacher Resource Bank.

	The Project Report Should Include	Amplification
1. Background and investigation	An introduction to the organisation.	This should include the type and purpose of the organisation, and give an idea of its size and scale of operation. The student's contact within the organisation should be identified.
	A description of the current system (or existing situation) and its environment.	Moving down in scale to describe the system (or existing situation) within the organisation/for the organisation, which is under investigation. The people/departments involved.
	Identification of client and users.	Who is it for? Clear identification of client (and users). N.B. It is likely that there is one client, but multiple users. At this level of study, it is expected that this will be the case.
	A business case (reasons) for change.	Why the project is needed by the organisation.
	Evidence of the use of relevant investigation techniques.	Evidence of, for example, planning, conducting, documenting and evaluating meetings with clients, interviews, observation, questionnaires, research as appropriate. (See the Teacher Resource Bank.)
	Requirements of the client.	What is the proposed system to provide?
2. Analysis and deliverables	Statement of scope.	This should include any internal or external constraints on the proposed system. These may include hardware, communication technologies, software, format of external information requirements, staffing and environmental factors.
	Description of the proposed system.	This may take various formats but should include the benefits for, and likely impacts on, the organisation.
	Documentation of processes.	Again, the format for the documentation is not fixed but should be appropriate to the system being analysed.
	Description of the users of the proposed system.	Details of the skills of the users of the proposed system, appropriate to the system.
	Evaluation criteria.	Both quantitative and qualitative criteria against which the solution can be tested and evaluated. These should be derived from the requirements of the client.

	The Project Report Should Include	Amplification
	Agreed deliverables.	What is to be produced and handed over to the client? This may be a prototype system or a partial system.
	Evidence of checking the findings with the client.	Findings must be presented in such a way as to be understandable to the client.
3. Design and planning for implementation ¹	Evidence of investigating alternative design solutions.	Should include investigating options for all elements of an ICT system as appropriate to the project.
	Draft design work.	To be able to discuss with client.
	Final design work.	From which to implement.
	Plan for implementation, testing and instalment, including proposed time scales.	
	Training requirements for the new system.	This may include what documentation will be required for training the users.
	Testing strategy.	Should set out what testing is necessary, who will do it, when and where. Students should describe here any constraints on live testing that require a simulated environment to be used.
	Test plan.	The tests that will be undertaken, what the tests are testing, the order in which the tests will be completed.
4. Testing and documentation of the implementation	Evidence of testing.	Where testing in the intended environment is not possible, testing in a simulated environment will be acceptable providing this has been explained and justified in the testing strategy. Testing should concentrate on the testing of complete processes and the system as a whole. Evidence of functional or unit tests, for example for validation or for the backing up of one file, is not required.
	Evidence of testing should include evidence of client and/or end user testing.	As appropriate, depending on whether the client is also the end user.
	Comprehensive documentation of the solution that would allow the solution to be used/maintained or developed further which is appropriate for the client/users.	Documentation could be technical or user documentation depending on the individual project undertaken and the agreed deliverables.
5. Evaluation of the implemented solution	A critical evaluation of the solution that would allow the solution to be used/maintained or developed further which is appropriate for the client/user.	The solution should be critically evaluated against the evaluation criteria and the client needs, as defined during the investigation and analysis stages. Evidence from testing should be referenced.
	An evaluation of the student's own performance.	This should identify strengths and weaknesses in the approach they have identified, how they would improve their performance on a similar project in the future.

N.B. Students are not required to produce separate documentary evidence of implementation as this is provided through testing evidence and the final documentation of the solution.

¹ The verb 'to implement' is taken to mean to create or produce an ICT-related system. Thus 'implementation' refers to whatever has been created.

	The Project Report Should Include	Amplification
6. The Project Report	The complete work should be submitted in the format of a project report.	The report should be well structured and should make use of facilities available within software packages to enable this to be done: for example, by including a contents list, headers/footers, pagination, indexes, effective use of appendices and presentation techniques. Appropriate use of technically accurate illustrative and visual material for effective communication. An appropriate style of writing has been adopted. The standard of the student's written communication will be assessed in the report.

Marking Criteria

1	Background and investigation	14 marks
2	Analysis and deliverables	15 marks
3	Design and planning for implementation	14 marks
4	Testing and documentation of implementation	13 marks
5	Evaluation of the implemented solution	7 marks
6	The Project Report	7 marks
	Total	70 marks

1 Background and investigation (14 marks)

This section is intended to assess the candidate's ability to investigate situations, work with others, use appropriate techniques and analysis, and present their findings in an appropriate manner.

0 - 6 MARKS	7 - 10 MARKS	11 - 14 MARKS
The candidate has provided a brief introduction to the organisation and the current system (or existing situation), showing some understanding. The client for the ICT-related system has been identified and the candidate shows that he/she has an idea of why the new system is required and what the client requires. The candidate has shown that he/she can use one investigation technique.	The candidate has provided an introduction to the organisation and shows understanding of the current system (or existing situation). The client for the ICT-related system and some of the users have been identified and the candidate shows that he/she has some understanding of why the new system is required. The client's requirements have been listed. The candidate has shown that he/she can use more than one investigation technique with an understanding of why it is appropriate for the task.	The candidate's introduction to the organisation and description of the organisation are comprehensive, showing a full understanding of the system, the organisation and its environment. The candidate has described the client and all users of the proposed system (both technical and non-technical, if appropriate). A justification of the required system is given with full understanding shown of the client's requirements. The candidate has used more than two investigation techniques appropriate for the task.

2 Analysis and deliverables (15 marks)

This section is intended to assess the candidate's ability to think analytically and logically, to present their findings using appropriate techniques, and to produce a specification to use during Design.

0 - 5 MARKS	6 - 9 MARKS	10 - 15 MARKS
A simple problem has been analysed with limited scope. The candidate has produced a brief statement of the scope of the project and has included a superficial description of the proposed system that shows some ability to analyse a system. Processes have been identified and some understanding of what is involved in the new system is shown. The candidate has attempted to identify the skills of a user or some users. Evaluation criteria have been provided, showing some understanding of the need for the objective assessment of a solution. Some deliverables have been referred to in the report, showing that the candidate has an idea of what he/she has to produce. (Note: the maximum mark achievable in this section is 5 if a simple task, rather than a system, has been analysed.)	A substantial problem has been analysed. There is a statement of the scope of the project. The candidate shows a good understanding of the proposed system and the likely benefits. The candidate has outlined the processes in the new system, showing some understanding of what is involved in each process. The skills of all users have been described, showing an understanding of the need to consider user skills when designing a solution. Both qualitative and quantitative evaluation criteria have been included, showing some understanding of the need for the objective assessment of a solution. The deliverables are listed, demonstrating some understanding of what the candidate has to produce. There is some evidence that findings have been checked with the client, demonstrating that the candidate understands the need for client involvement.	A substantial problem has been analysed. There is a statement of scope along with an appreciation of any internal or external constraints on the project. The candidate has described the proposed system and included the benefits and likely impacts on the organisation. The candidate shows an understanding of the processes, which are fully documented and include the inputs and outputs required. The documentation could be used for design work, the candidate exhibiting strong skills of analysis and logical thinking. The skills of all users have been described, showing an understanding of the need to consider user skills in designing a solution. How the users will use the proposed system is described. Qualitative and quantitative evaluation criteria have been included that match the client requirements and show a good understanding of the need for the objective assessment of a solution. The candidate shows a full understanding of the deliverables that are required. There is evidence of how all findings have been checked with the client and amended as necessary, demonstrating an effective response and full understanding of the need for client involvement.

3 Design and planning for implementation (14 marks)

This section is intended to assess the candidate's ability to design and document effective solutions, using their capacity to think creatively and innovatively, and to select appropriate solutions. A candidate's skills in logical thought and planning are also assessed through his/her ability to design appropriate implementation and testing strategies and plans.

0 - 5 MARKS	6 - 10 MARKS	11 - 14 MARKS
The candidate has shown evidence of investigating alternative solutions. The candidate has included design work for the new system. The candidate has produced a limited plan for testing and installing the system with suggested deadlines, showing a basic understanding of the issues involved. The training required has been identified. The candidate shows limited understanding of the testing process, but has included a test plan that may only include functional testing. (Note: the maximum mark achievable in this section is 5 if a simple task, rather than a system, has been designed and planned.)	The candidate understands the basic principles of investigating and evaluating alternative solutions. The candidate has shown evidence of understanding the iterative nature of the design process by including design work for the new system that he/she has amended as a result of client feedback. Training needs have been considered and training is considered in a time plan that shows expected durations as well as deadlines, showing some understanding of the need to quantify time requirements. There is a test strategy for testing the effectiveness of the implemented system. The test plan shows an understanding of the need to test the system against evaluation criteria or client needs.	Alternative solutions have been considered and evaluated objectively in relation to client needs. The candidate shows full understanding of the design process and has included draft design work, which has been amended as a result of well-evidenced client feedback, and the resultant final design for the new system. Training needs have been considered and training has been included in a time plan that shows expected durations as well as deadlines, showing understanding of the need to quantify time requirements. The process of testing a system is well understood with a strategy and plan that includes testing against both evaluation criteria and client needs.

4 Testing and documentation of the implementation (13 marks)

This section is intended to assess the candidate's ability to implement and document effective solutions, shown through his/her testing and documentation of the system.

0 - 5 MARKS	6 - 9 MARKS	10 - 13 MARKS
The candidate has carried out some testing of the implementation that shows a limited understanding of the testing process. Some documentation has been provided that partly describes the implemented solution. The documentation would not be easily understood by a third party. (Note: the maximum mark achievable in this section is 5 if a simple task, rather than a system, has been tested and documented.)	There is evidence that the candidate understands the need to test the processes in the system and the need to involve the client in the testing process. The candidate has shown client involvement in testing. The documentation is appropriate and comprehensive. The candidate has made some attempts to make the documentation suitable for the client and/or user(s) and, as a result, the documentation is understandable.	Appropriate testing of the whole system has been carried out, showing understanding of the purpose of testing. The candidate understands the need to involve both the client and user(s) in the testing process and has explained how he/she has achieved this, with evidence being shown. Documentation is comprehensive and could easily be used by the client and/or user(s), and for maintenance or upgrading of the solution – thus showing a good understanding of the purpose of documentation for an ICT-related system.

5 Evaluation of the implemented solution (7 marks)

This section is based upon a candidate's ability to carry out logical, reasoned, critical evaluation that is related to the evaluation criteria and client's needs, for the solution, and to his/her own development.

0 - 4 MARKS	5 - 7 MARKS
The candidate shows skills in evaluation by carrying out an evaluation of his/her implemented solution. Some strengths or weaknesses in the solution have been referred to. Reference is made in the evaluation to the client's needs or the evaluation criteria. The candidate shows some understanding of the need to evaluate his or her own performance in completing the work.	Evaluation is well understood, with critical evaluation having been completed that identifies strengths and weaknesses of the solution and suggests possible improvements that could be made. Evaluation refers to the client's needs and evaluation criteria. The candidate has identified the strengths and weaknesses in the approach he/she has taken to his/her work and has identified how he/she could improve his/her performance on similar work in the future.

6 The Project Report (7 marks)

The quality of written communication and the ability to use appropriate documentation, as well as the software skills needed to create it, are assessed in this section.

0 - 4 MARKS	5 - 7 MARKS
The candidate shows some understanding of the requirements of a professional report. Software features have been used appropriately. The language used by the candidate is generally appropriate to the report, but the content may be disorganised. Some illustrative material has been included, but shows that the candidate has only a basic understanding of how to use illustrative material appropriately.	A well-structured report has been produced to a professional standard that makes good use of the features available in the software package used. The language used is suitable and the report is logical and well organised. The report is well illustrated, with appropriate material showing that the candidate has an understanding of the use of such material.